

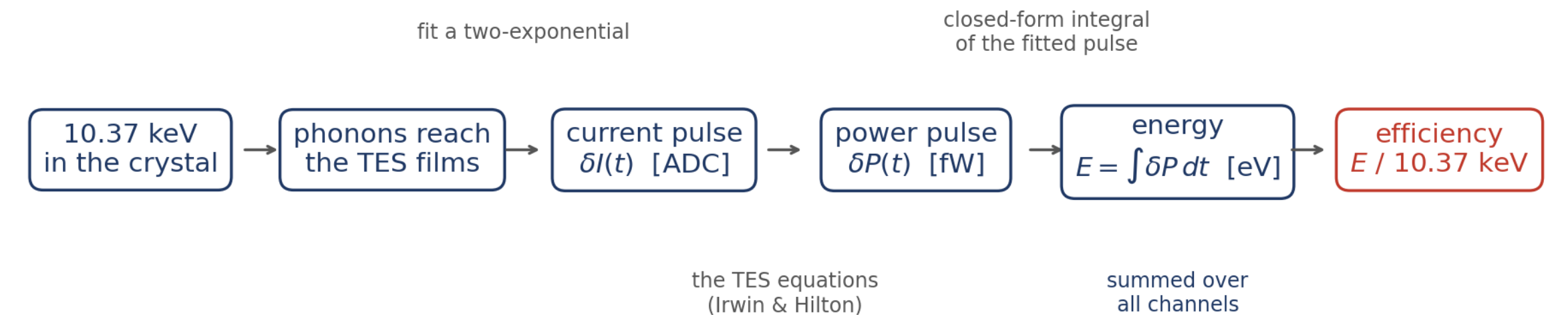
Phonon collection efficiency of Z7

Z7 的声子收集效率

Zhiheng Li · University of Minnesota · SuperCDMS SNOLAB Run 4 · September 2026
github.com/ZhihengLi0/urop → snolab/collection_efficiency/

李知恒 · 明尼苏达大学 · SuperCDMS SNOLAB Run 4 · 2026 年 9 月
github.com/ZhihengLi0/urop → snolab/collection_efficiency/

The question

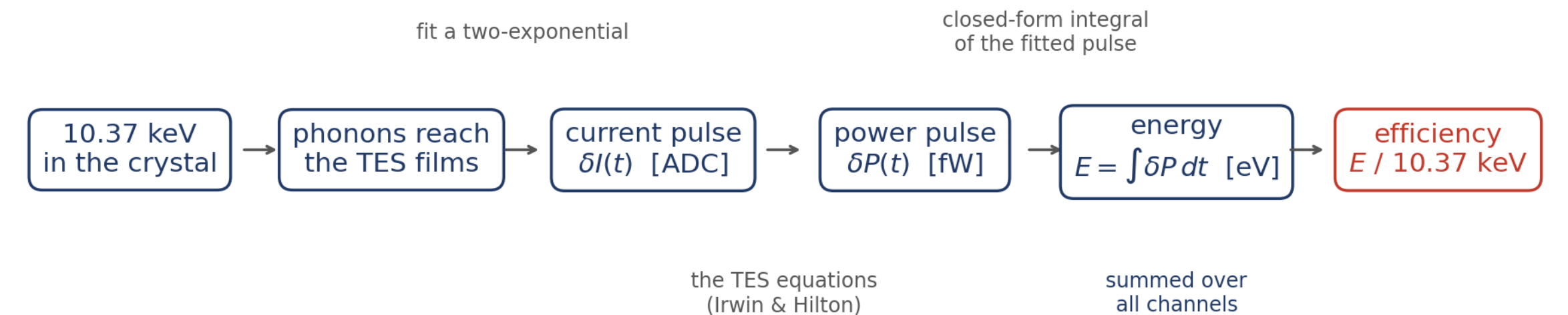


For detectors operated at 0 V (no NTL effect):

Collection efficiency = energy absorbed by the TESs, summed over all channels ÷ the 10.37 keV of the event

Sample: Z7, the K-line events of the Ge activation run, from SNOLAB R4 (2207 events, 30 series, all 11 channels).

问题是什么



对于在 0 V 下工作的探测器（没有 NTL 效应）：

收集效率 = TES 吸收的能量（所有通道求和） ÷ 事件放进晶体的 10.37 keV

样本：Z7，Ge 活化的 K 线事件，来自 SNOLAB R4（2207 个事件，30 个 series，11 个通道）。

The formula, and the two numbers it needs

Method 1 of the CDMS collection-efficiency note

$$\delta P(t) = I_0 (R_L - R_0) \delta I(t) + 2 R_L (\delta I(t))^2$$

linear term

quadratic term

$$E = \int \delta P(t) dt = I_0 (R_L - R_0) A (\tau_f - \tau_r) + 2 R_L A^2 \left(\frac{\tau_f + \tau_r}{2} - \frac{2 \tau_f \tau_r}{\tau_f + \tau_r} \right)$$

for the fitted pulse $\delta I(t) = A [e^{-t/\tau_f} - e^{-t/\tau_r}]$ integrated from $-\infty$ to $+\infty$: zero before it starts, decays to zero after

Z7 PBS1 bias point (measured, from detectorConfig): $I_0 = -18.07 \mu A$, $R_0 = 44.16 \text{ m}\Omega$, $R_L = R_p + R_{sh} = 19.24 \text{ m}\Omega$

$$\Rightarrow I_0 (R_L - R_0) = 0.45 \mu V \quad \text{and} \quad 2 R_L = 0.0385 \Omega$$

δP = the change of Joule heating when the current moves by δI : a linear term and a small quadratic term.

The note also gives Method 2. In our sign convention it reads $\delta P = I_0 (R_L - R_0) \cdot \delta I - R_L \cdot (\delta I)^2$: the same linear term, only the quadratic term differs, and it gives 1.7 % less energy (PBS1, median of 15 events).

公式和它需要的两个数

采用 CDMS 收集效率文档里的 Method 1

$$\delta P(t) = I_0 (R_L - R_0) \delta I(t) + 2 R_L (\delta I(t))^2$$

linear term

quadratic term

$$E = \int \delta P(t) dt = I_0 (R_L - R_0) A (\tau_f - \tau_r) + 2 R_L A^2 \left(\frac{\tau_f + \tau_r}{2} - \frac{2 \tau_f \tau_r}{\tau_f + \tau_r} \right)$$

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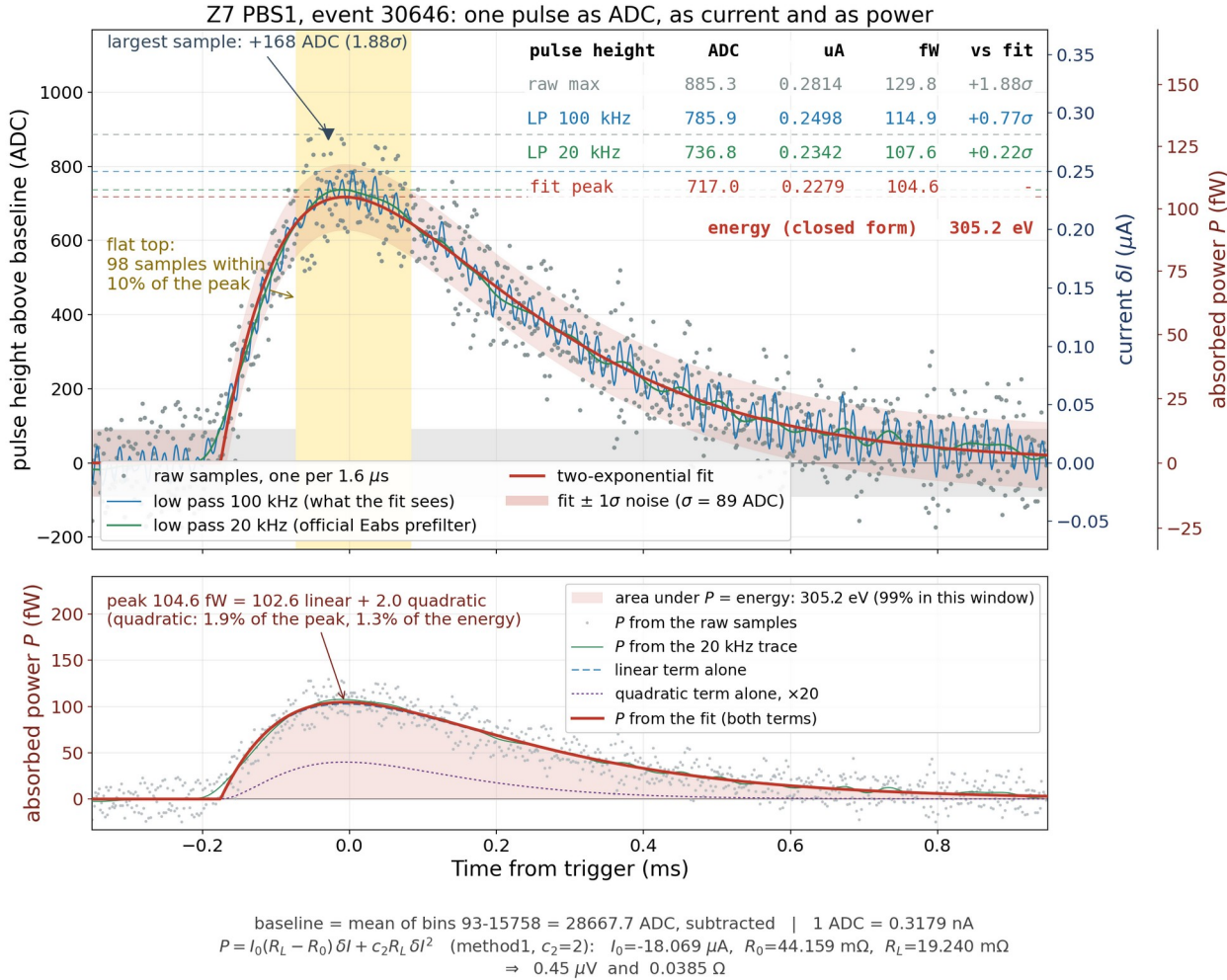
$$\Rightarrow I_0 (R_L - R_0) = 0.45 \mu V \quad \text{and} \quad 2 R_L = 0.0385 \Omega$$

δP = 电流变化 δI 之后焦耳热的变化量：一个线性项，加一个很小的二次项。

文档里还有一个 Method 2。换成我们的符号约定是 $\delta P = I_0 (R_L - R_0) \cdot \delta I - R_L \cdot (\delta I)^2$ ：线性项相同，只有二次项不同，能量低 1.7%（PBS1，15 个事件的中位数）。

Example: one pulse

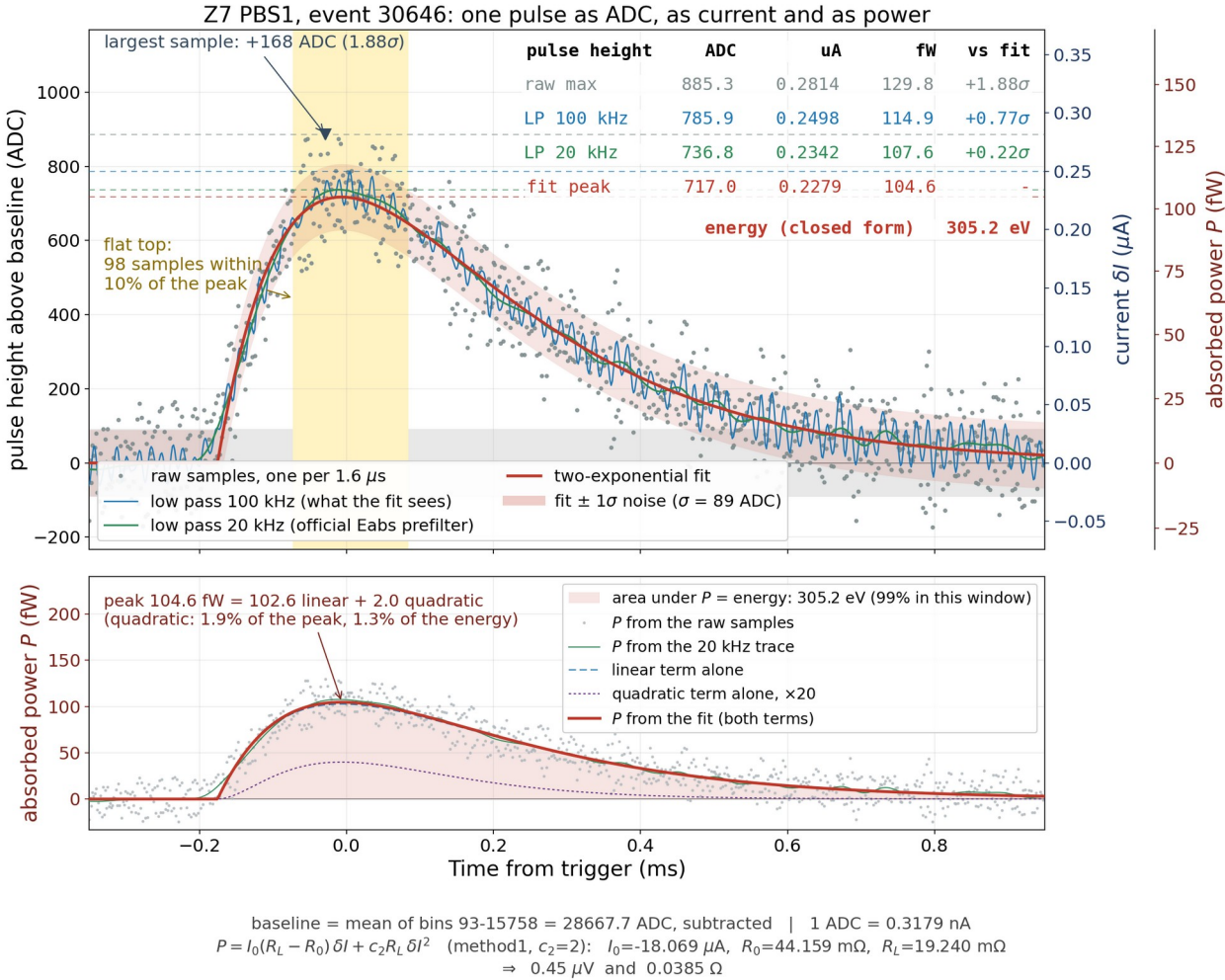
Z7 PBS1, event 30646: the same trace as ADC counts, as current, as power



The two-exponential is fitted to the 100 kHz low-passed trace (blue), not to the raw samples; the raw samples are only drawn.

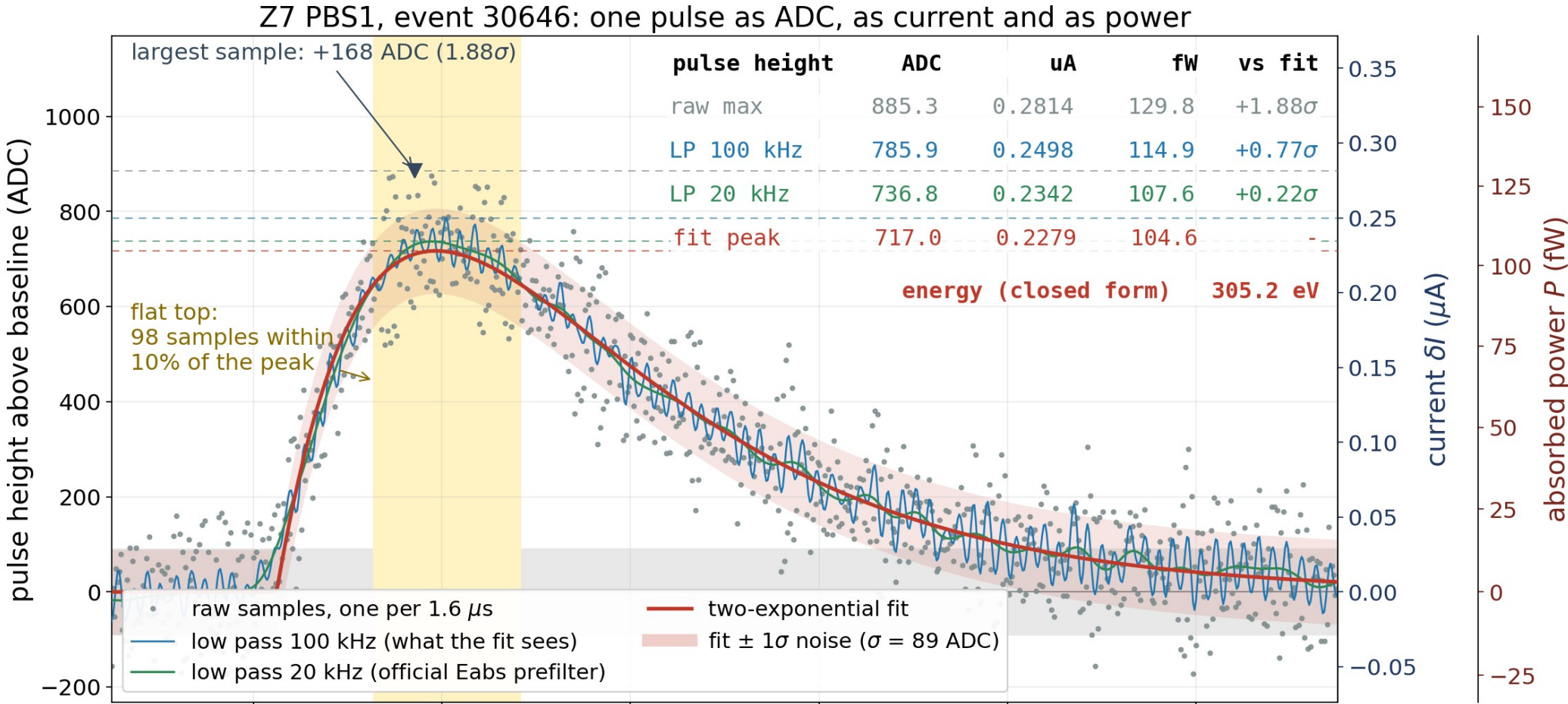
例子：一个脉冲

Z7 PBS1 , 事件 30646 : 同一条波形，读成 ADC 计数、读成电流、读成功率率



双指数拟合用的是 100 kHz 低通之后的波形（蓝线），不是原始采样点；原始采样点只是画出来对照。

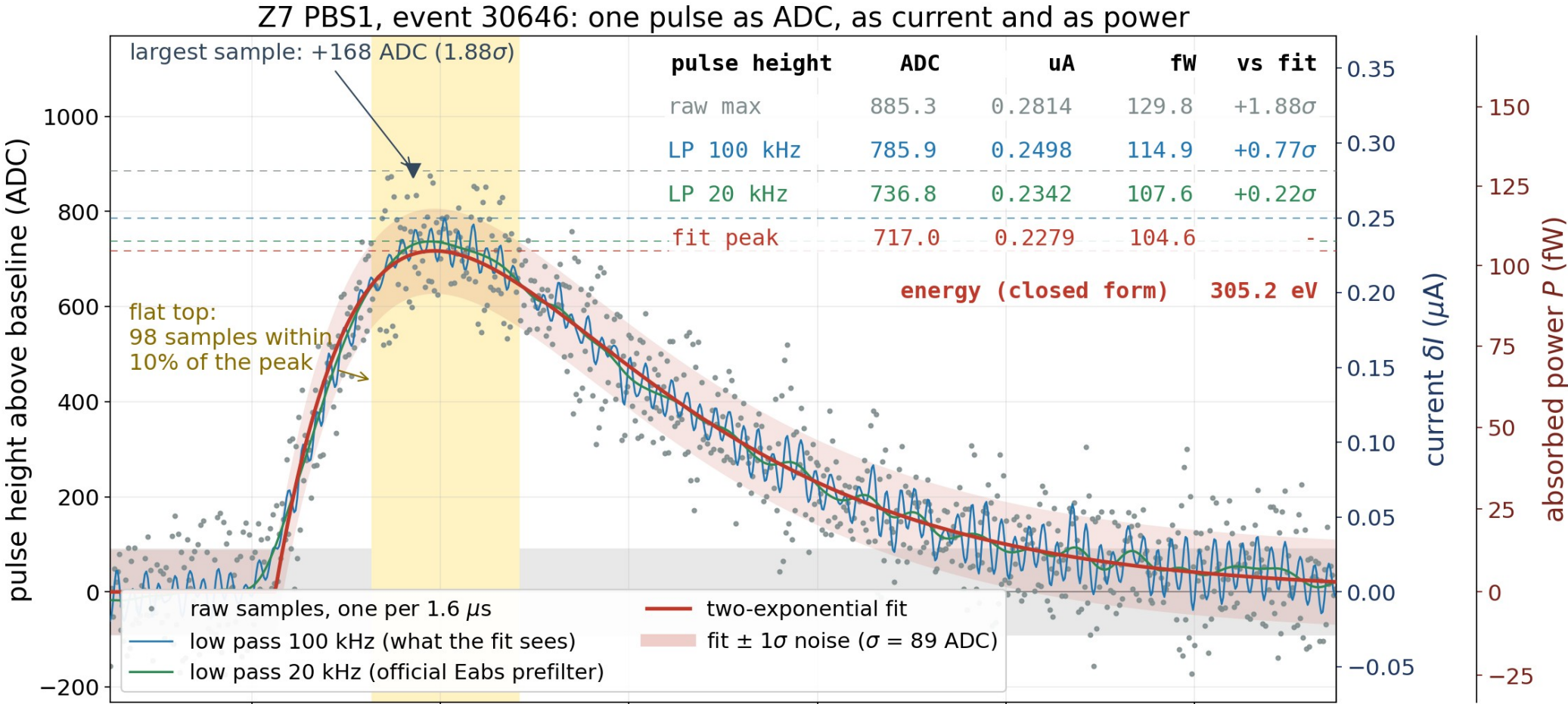
The height: take it from the fit, not from the largest sample



Yellow band: 98 samples sit within 10 % of the peak. The largest of them is the largest noise excursion.

Largest raw sample 885 ADC, fit 717 ADC: the maximum reads 23 % high. Filtering helps; the fit removes it.

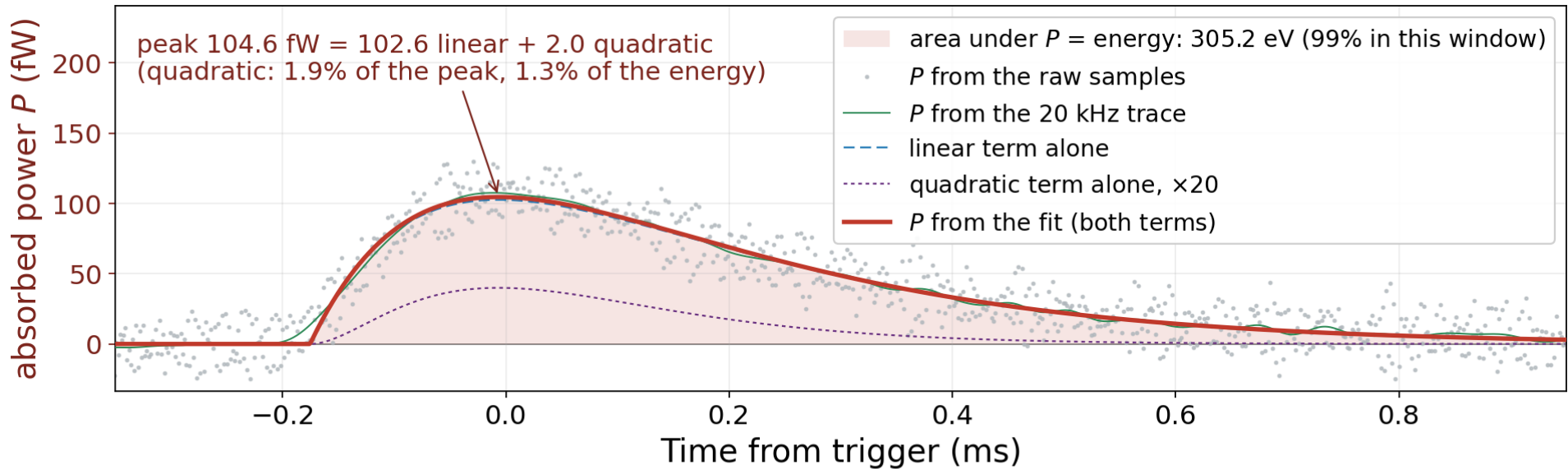
脉冲高度：从拟合取，不从最大的采样点取



黄色竖带：98 个采样点都在峰高的 10% 以内。其中最大的那个，就是噪声往上跳得最多的那个。

原始数据最大值 885 ADC，拟合峰值 717 ADC：取最大值偏高 23%。滤波能减轻，拟合能消除。

The power pulse, and its area is the energy



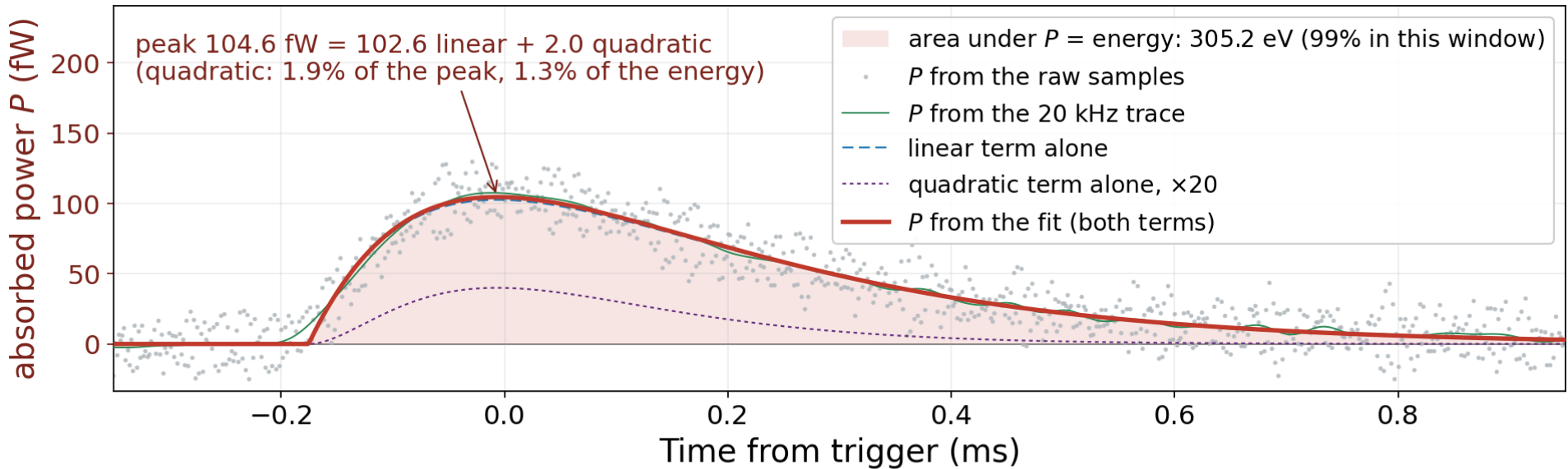
baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA
 $P = I_0(R_L - R_0) \delta I + c_2 R_L \delta I^2$ (method1, $c_2=2$): $I_0=-18.069 \mu\text{A}$, $R_0=44.159 \text{ m}\Omega$, $R_L=19.240 \text{ m}\Omega$
 $\Rightarrow 0.45 \mu\text{V}$ and 0.0385Ω

Red: the fitted current put through the formula. Blue dashed: the linear term alone. Purple: the quadratic term, $\times 20$.

The quadratic term is 1.9 % of the peak — the power pulse has the shape of the current pulse.

The shaded area is the energy this channel absorbed: 305 eV.

功率脉冲，它的面积就是能量



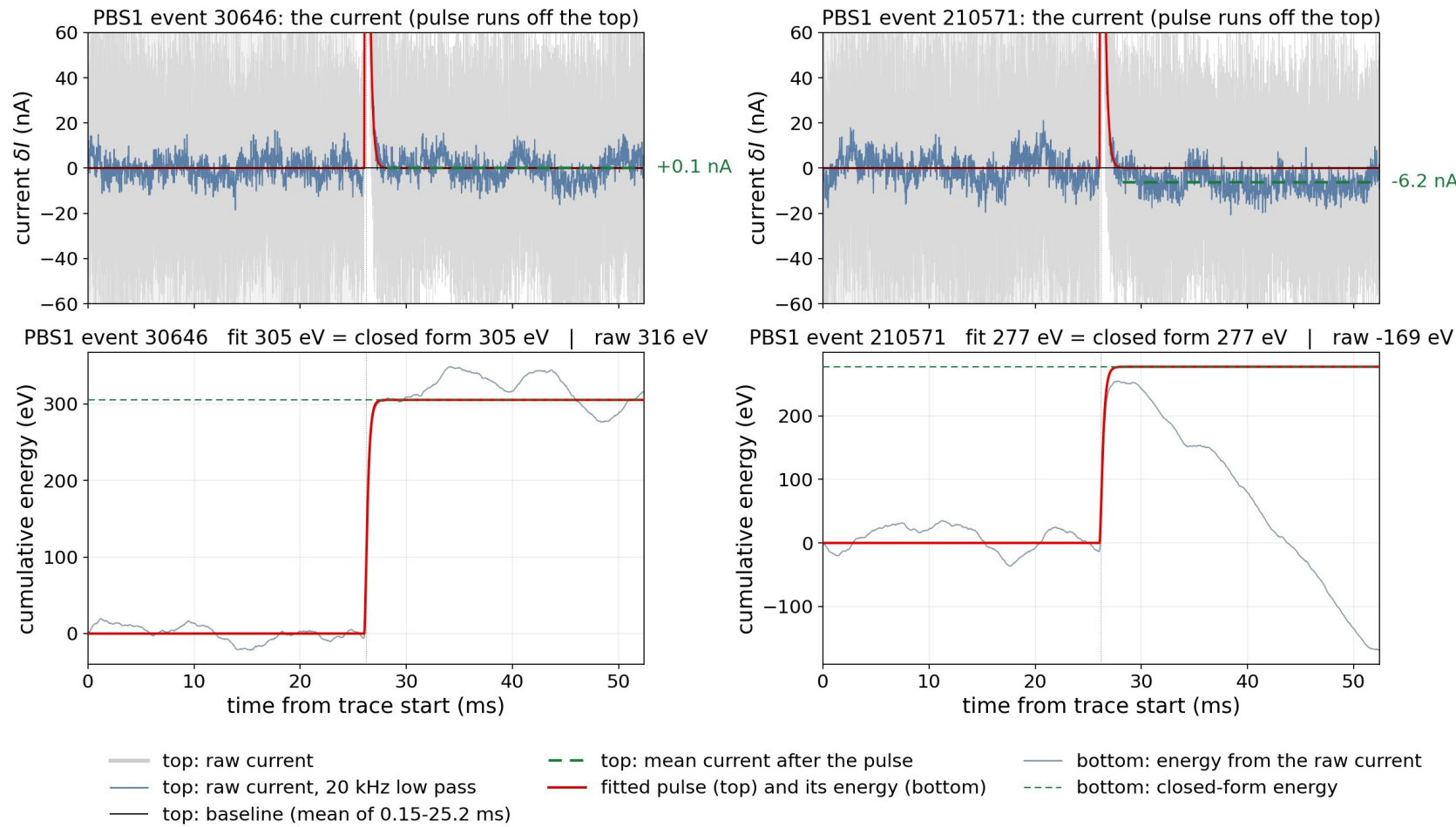
baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA
 $P = I_0(R_L - R_0) \delta I + c_2 R_L \delta I^2$ (method1, $c_2=2$): $I_0=-18.069 \mu\text{A}$, $R_0=44.159 \text{ m}\Omega$, $R_L=19.240 \text{ m}\Omega$
 $\Rightarrow 0.45 \mu\text{V}$ and 0.0385Ω

红线：拟合出的电流代进公式。蓝色虚线：只有线性项。紫色点线：只有二次项，放大 20 倍。

二次项只占峰值的 1.9% —— 功率脉冲的形状和电流脉冲一样。

粉色面积就是这个通道吸收的能量：305 eV。

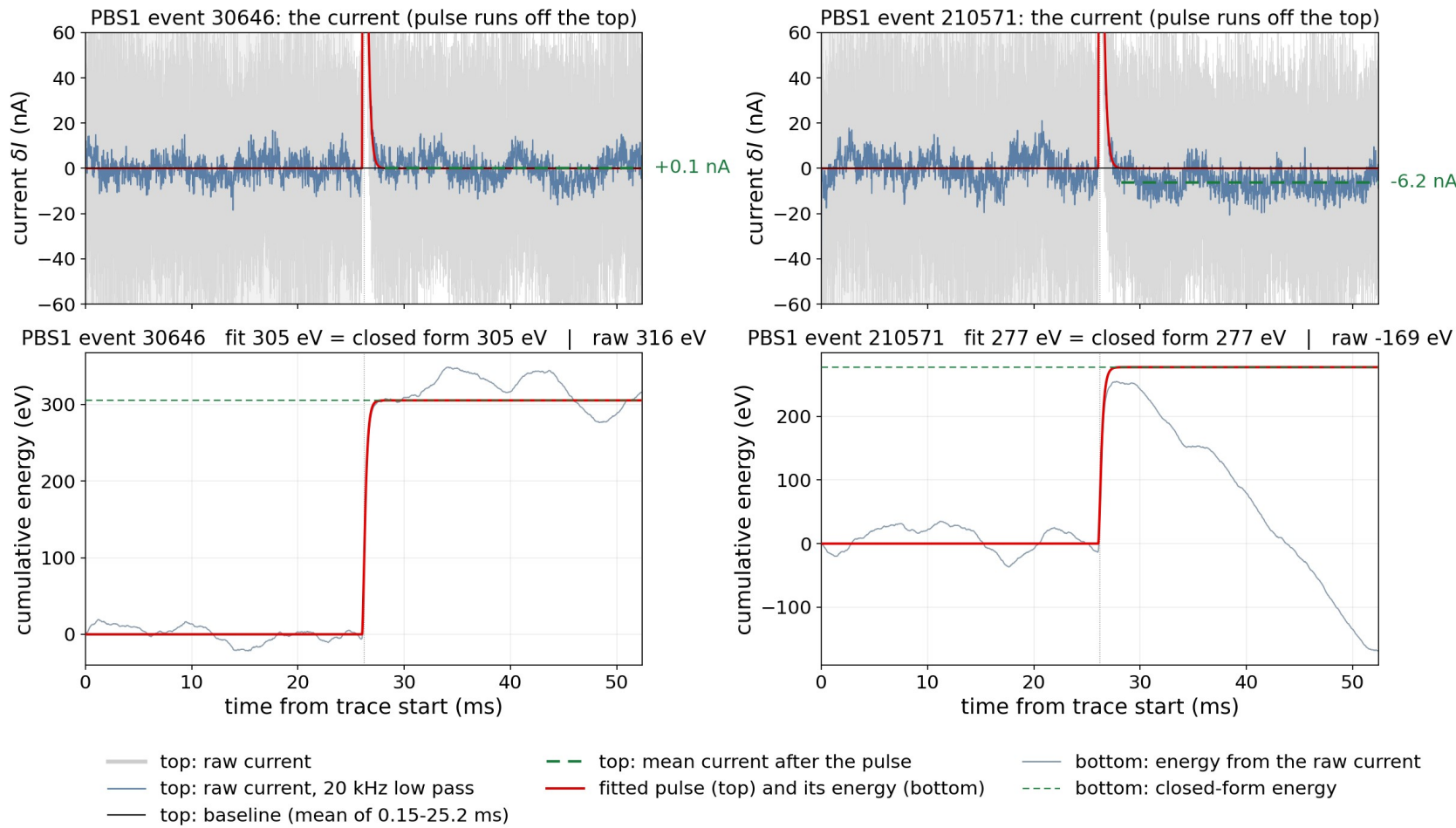
Why the fitted pulse is integrated, and not the data



Top: the raw current of the same two events, zoomed on the baseline. Bottom: the energy accumulated from it (grey) and from the fit (red).

Right: after the pulse the current sits 6 nA below its baseline — hidden in 28 nA of noise, yet over 24 ms it is -420 eV: raw -169 eV, fit 277 eV.

为什么积分的是拟合脉冲，不是原始数据

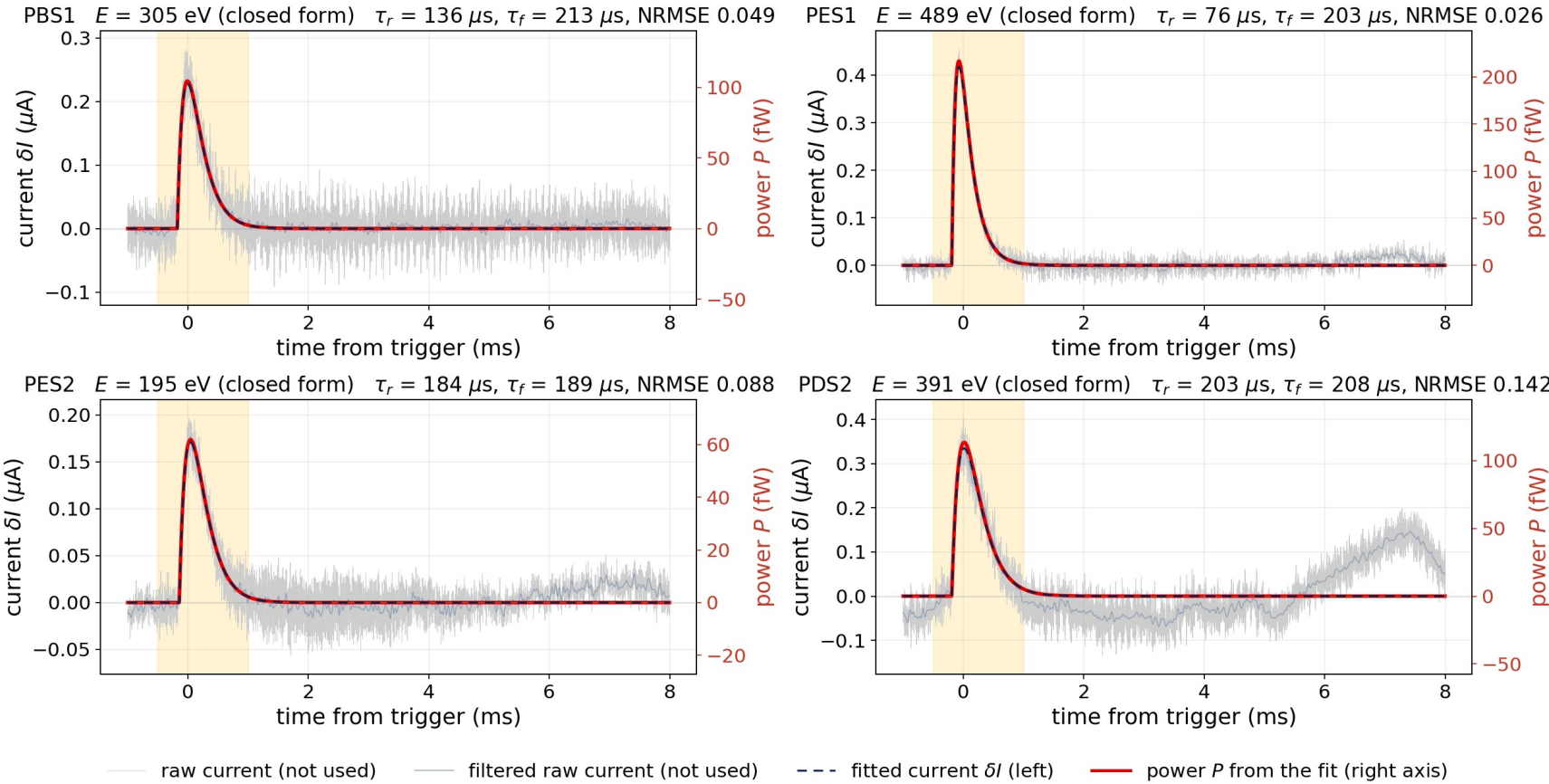


上：这两个事件的原始电流，纵轴放大到基线附近。下：由它累计的能量（灰）和由拟合累计的能量（红）。

右：脉冲过后电流比基线低 6 nA —— 埋在 28 nA 的噪声里看不出来，但 24 ms 积下来就是 -420 eV：原始 -169 eV，拟合 277 eV。

The same event in its other channels: they add up to 33 %

Z7 event 30646; four of the eleven channels

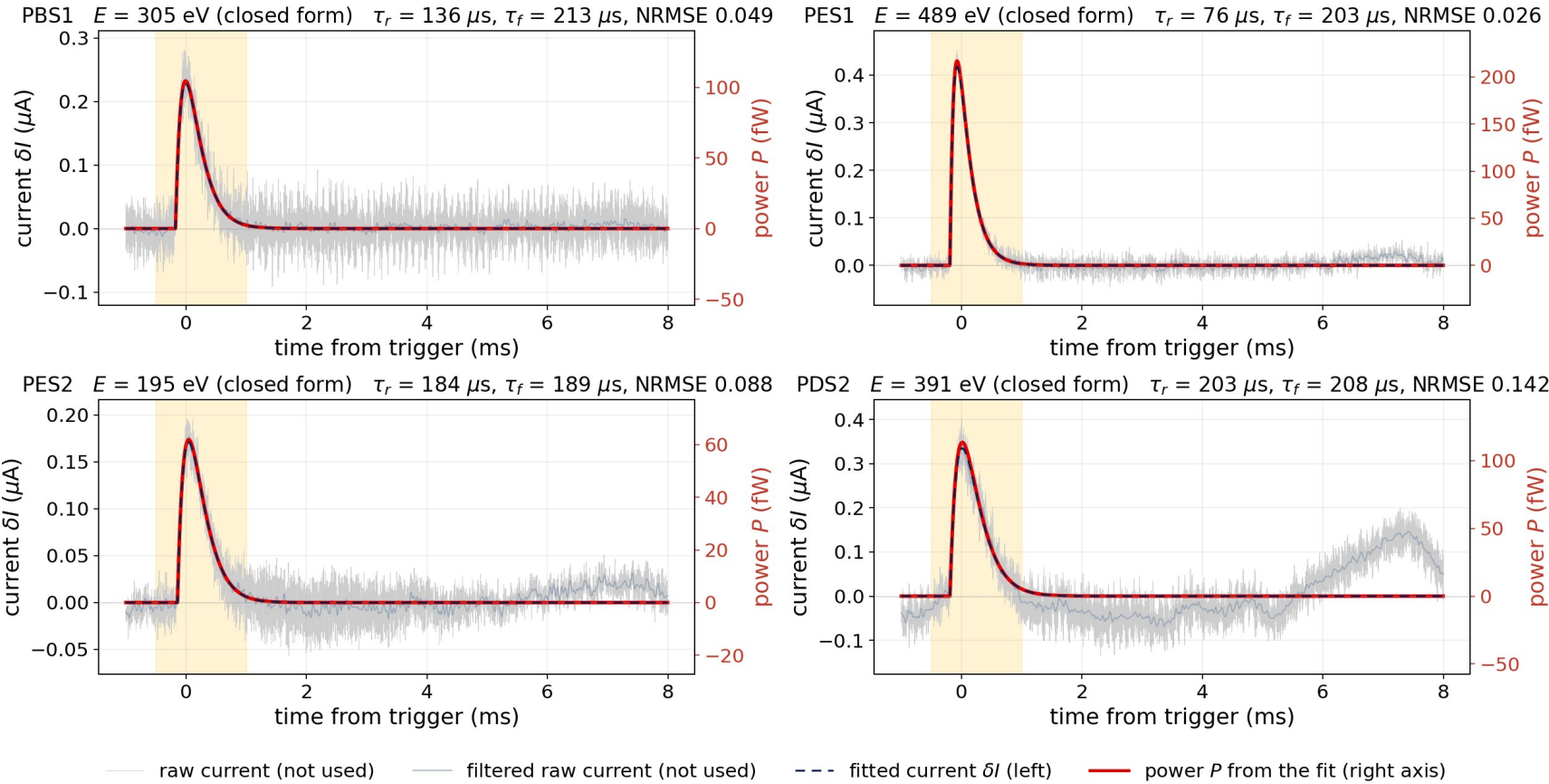


The channels share the energy unevenly (PES1 489 eV, PES2 195 eV): the event sat closer to one side.

Summed over the eleven channels: 3419 eV = 33.0 % of 10.37 keV. PDS2 (bottom right) carries a slow swing.

同一个事件的其他通道：加起来是 33%

Z7 事件 30646；11 个通道里的 4 个

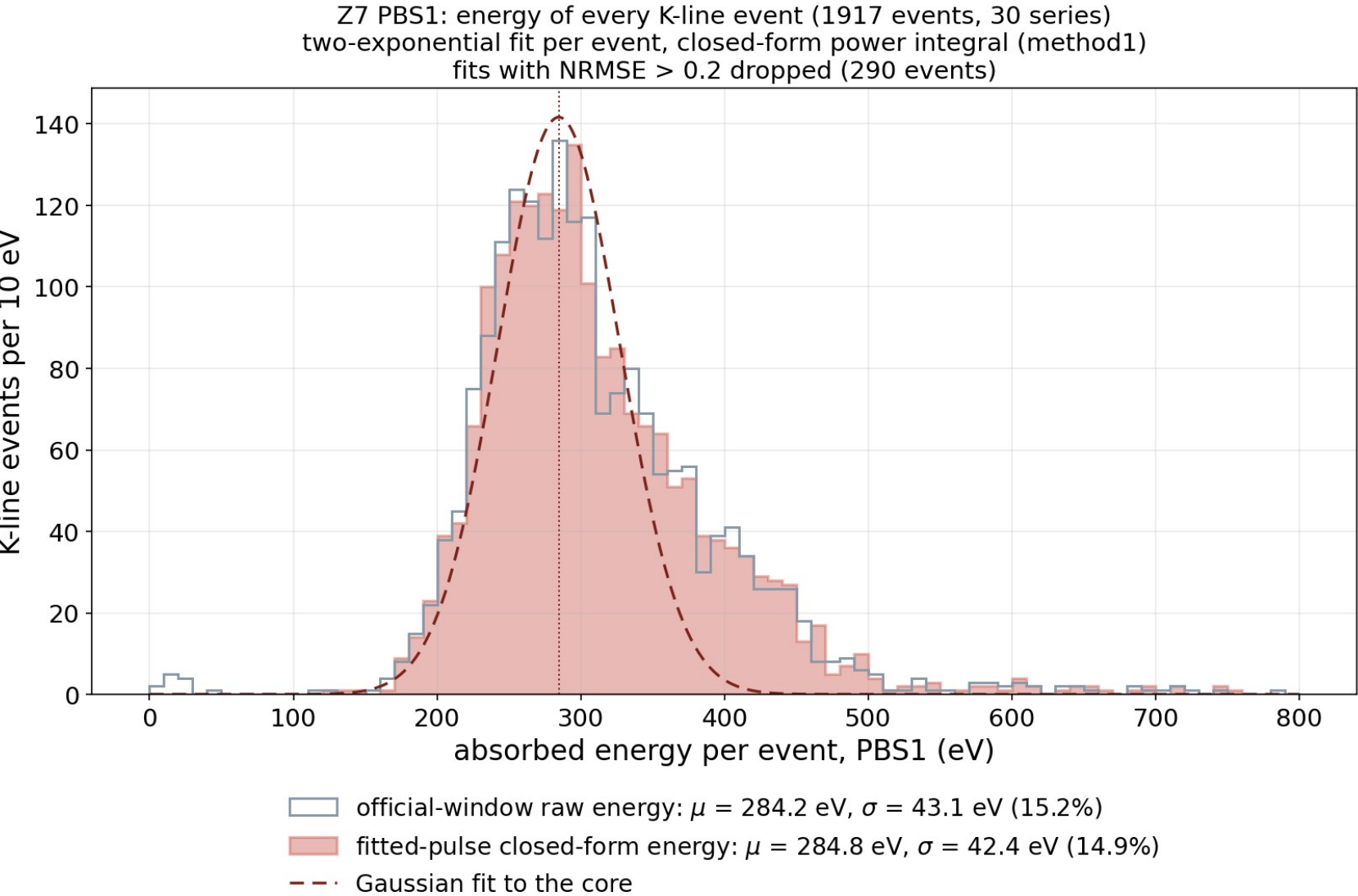


各通道分到的能量很不均匀（ PES1 489 eV ， PES2 195 eV ）：事件发生的位置更靠近一侧。

11 个通道加起来：3419 eV = 10.37 keV 的 33.0%。 PDS2（右下）有一个缓慢的大起伏。

Now every event: one channel gives a wide, lopsided distribution

PBS1, 1917 K-line events, one fit and one integral each



Red: from the fit. Grey: the official window on the raw trace. Same peak to 0.2 %.

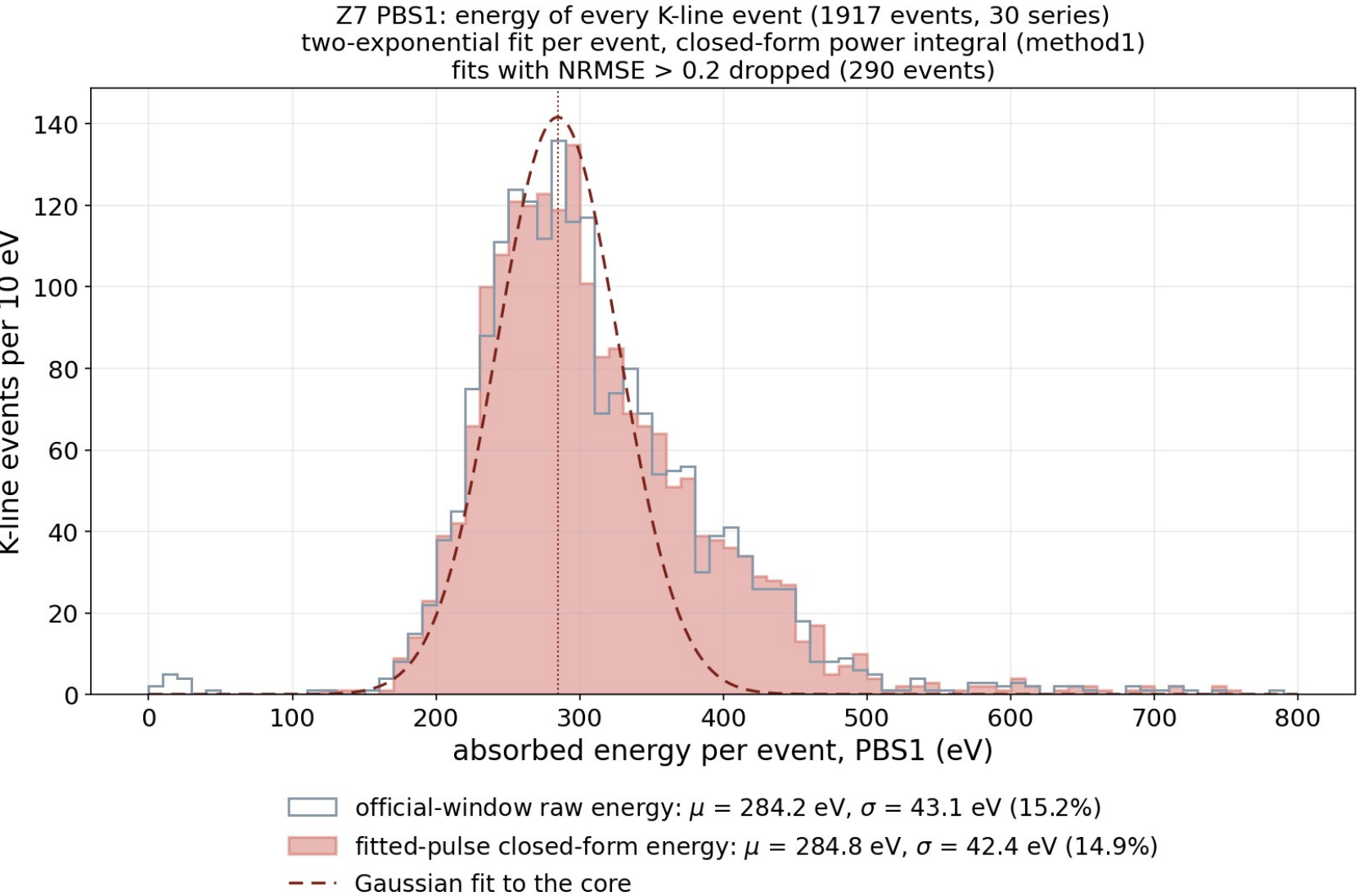
Peak 285 eV, width 15 % — for a line whose energy is fixed.

The tail to the right is the events that happened near this channel.

One channel's width is event position, not noise.

现在看所有事件：一个通道给出又宽又歪的分布

PBS1 ， 1917 个 K 线事件 ， 每个事件一次拟合、一次积分



红色：来自拟合。灰色：官方窗口在原始波形上算的。峰位只差 0.2 %。

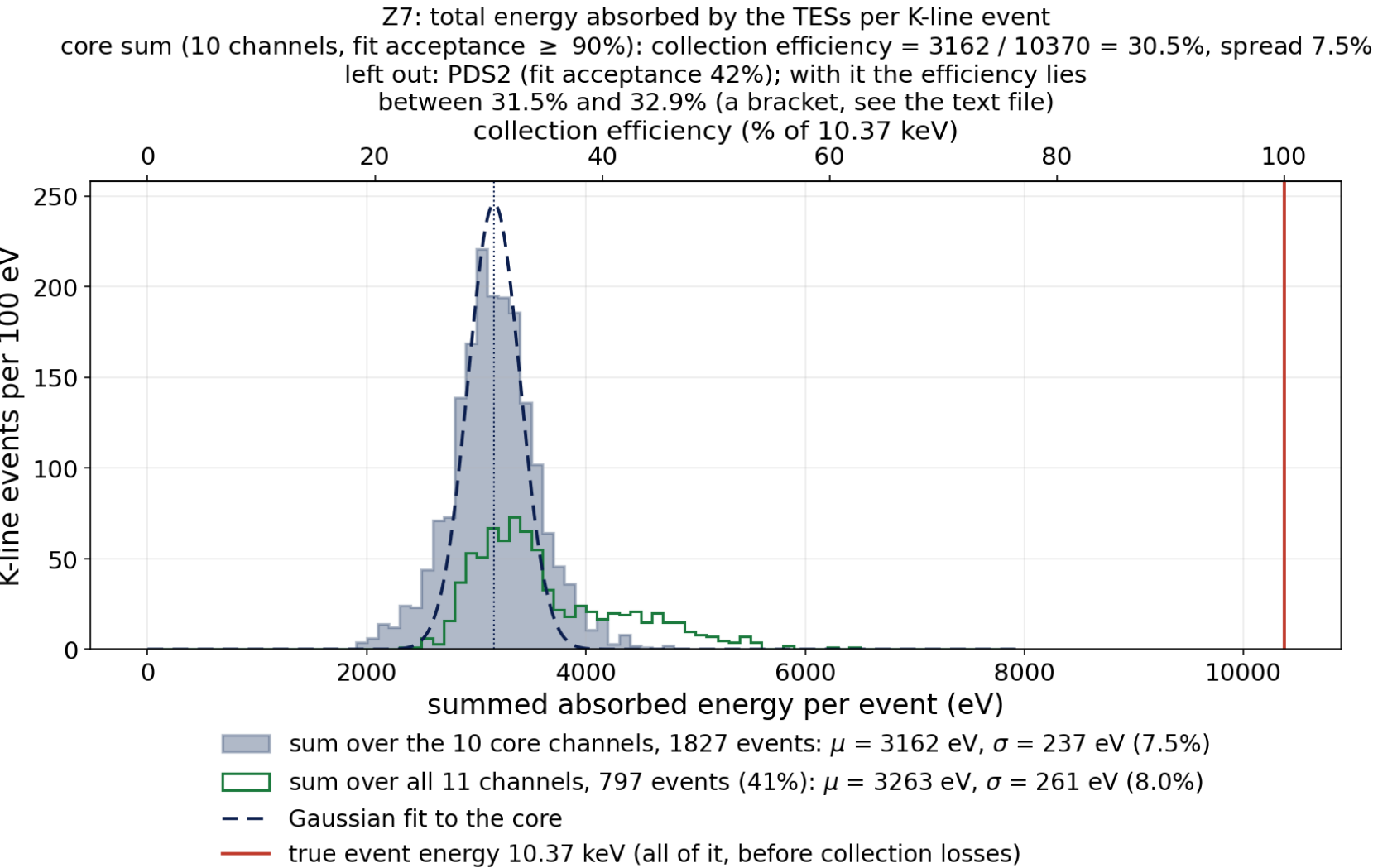
峰 285 eV ，宽度 15% —— 这可是一条能量固定的谱线。

往右拖的尾巴，是发生在这个通道附近的事件。

单通道的宽度是事件位置，不是噪声。

Summed over the channels: 32 ± 1 %

the total energy the TESs absorbed per K-line event



Ten channels summed, 1827 events: 3162 eV = 30.5 %, and the width drops to 7.5 % — the position effect cancels.

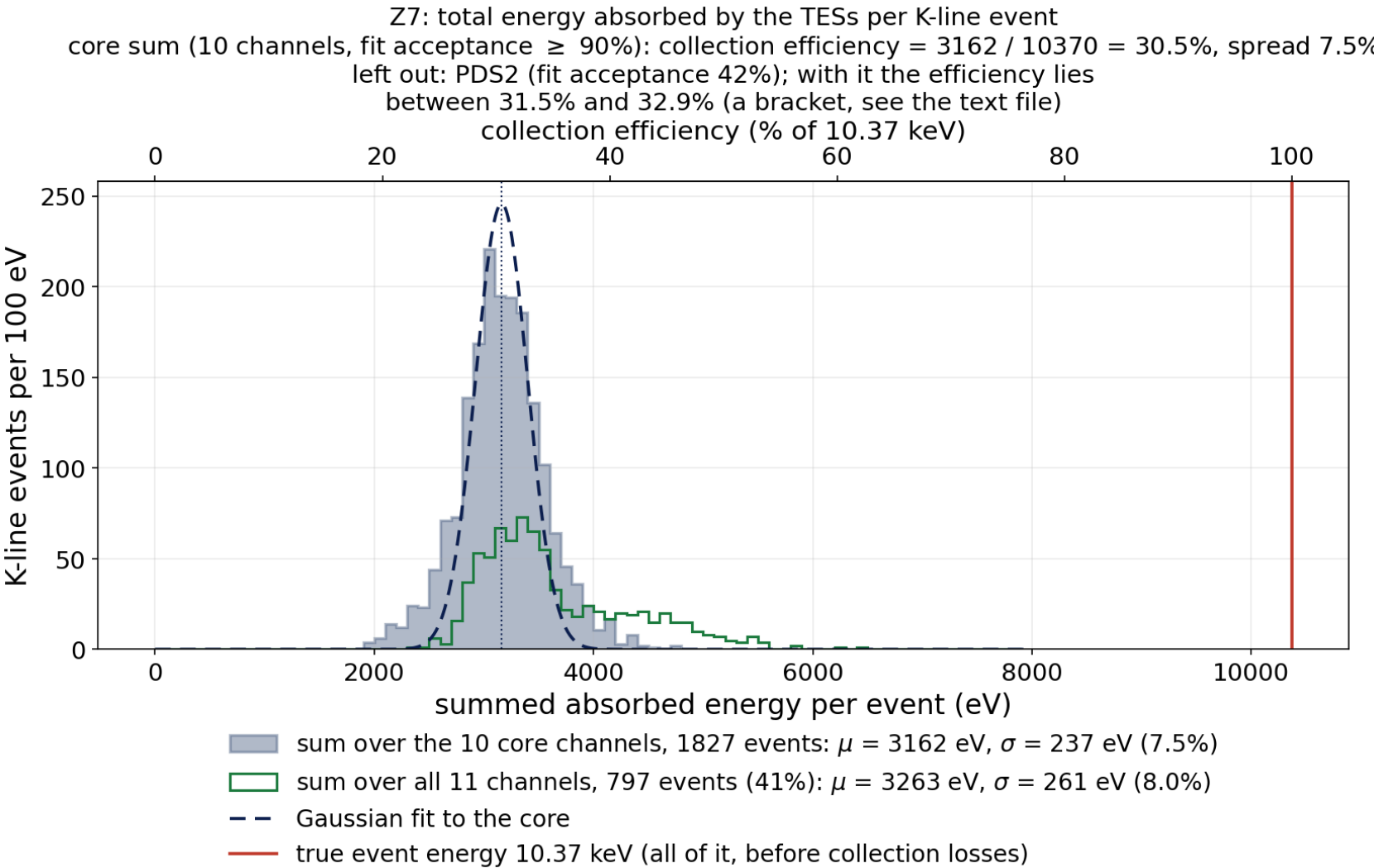
PDS2 fits only 42 % of the events (that slow swing), and those events are not typical; its share can only be bracketed.

Full efficiency 31.5–32.9 %: 32 ± 1 %.

Red line: 10.37 keV — two thirds of the energy never reaches the TESs.

各通道求和：32 ± 1%

每个 K 线事件被 TES 吸收的总能量



10 个通道求和，1827 个事件：3162 eV = 30.5%，宽度降到 7.5% —— 位置效应抵消了。

PDS2 只有 42% 的事件能拟合上（那个缓慢起伏），而且这些事件不典型；它的份额只能给一个范围。

完整效率 31.5–32.9%：32 ± 1%。

红线：10.37 keV —— 三分之二的能量根本没到 TES。

Summary, and what is still open

A current pulse becomes a power pulse through the TES equations; the area under the power is the energy. The quadratic term is 2 %.

The energy comes from the fitted pulse, integrated from $-\infty$ to $+\infty$: no bias from the noisy maximum, no drift.

One channel: 15 % wide and lopsided (position). Ten channels summed: 7.5 % and symmetric.

Z7 absorbs 32 ± 1 % of a 10.37 keV event in its TESs. The CDMS note on the R37 CUTE tower reports 26–41 %.

Open: Z7 only · the HV bias of Z7 is not established (0 V assumed) · no dI/dV, so the inductor term is dropped · PDS2 is the ± 1 % · 12.5 % of events fit in no channel and are not in the histograms.

总结，以及还没落实的

电流脉冲通过 TES 方程变成功率脉冲；功率下面的面积就是能量。二次项只有 2%。

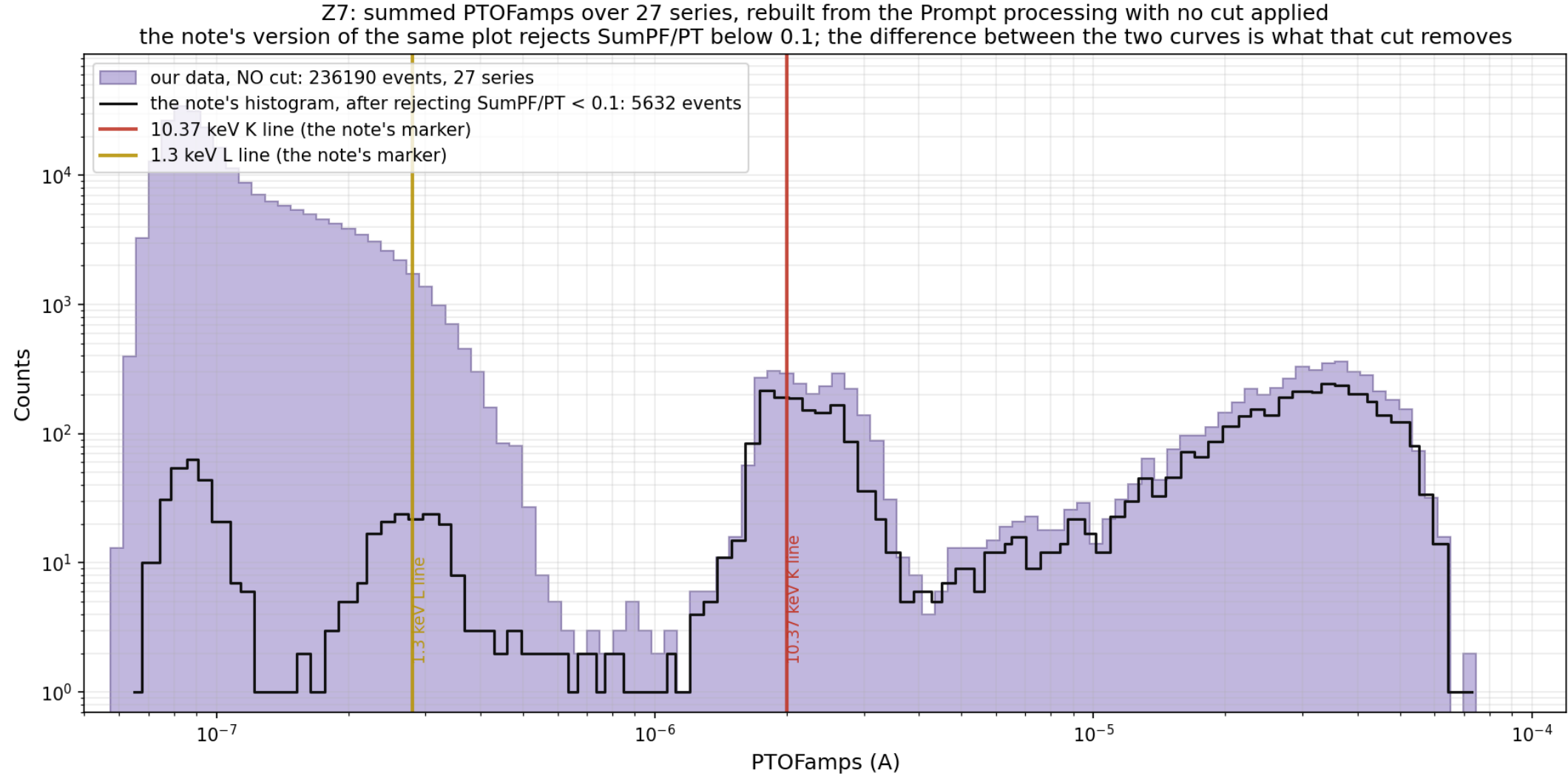
能量来自拟合脉冲、从 $-\infty$ 积到 $+\infty$ ：没有最大值的噪声偏差，没有漂移。

单通道：15% 宽、歪的（位置效应）。10 个通道求和：7.5%、对称。

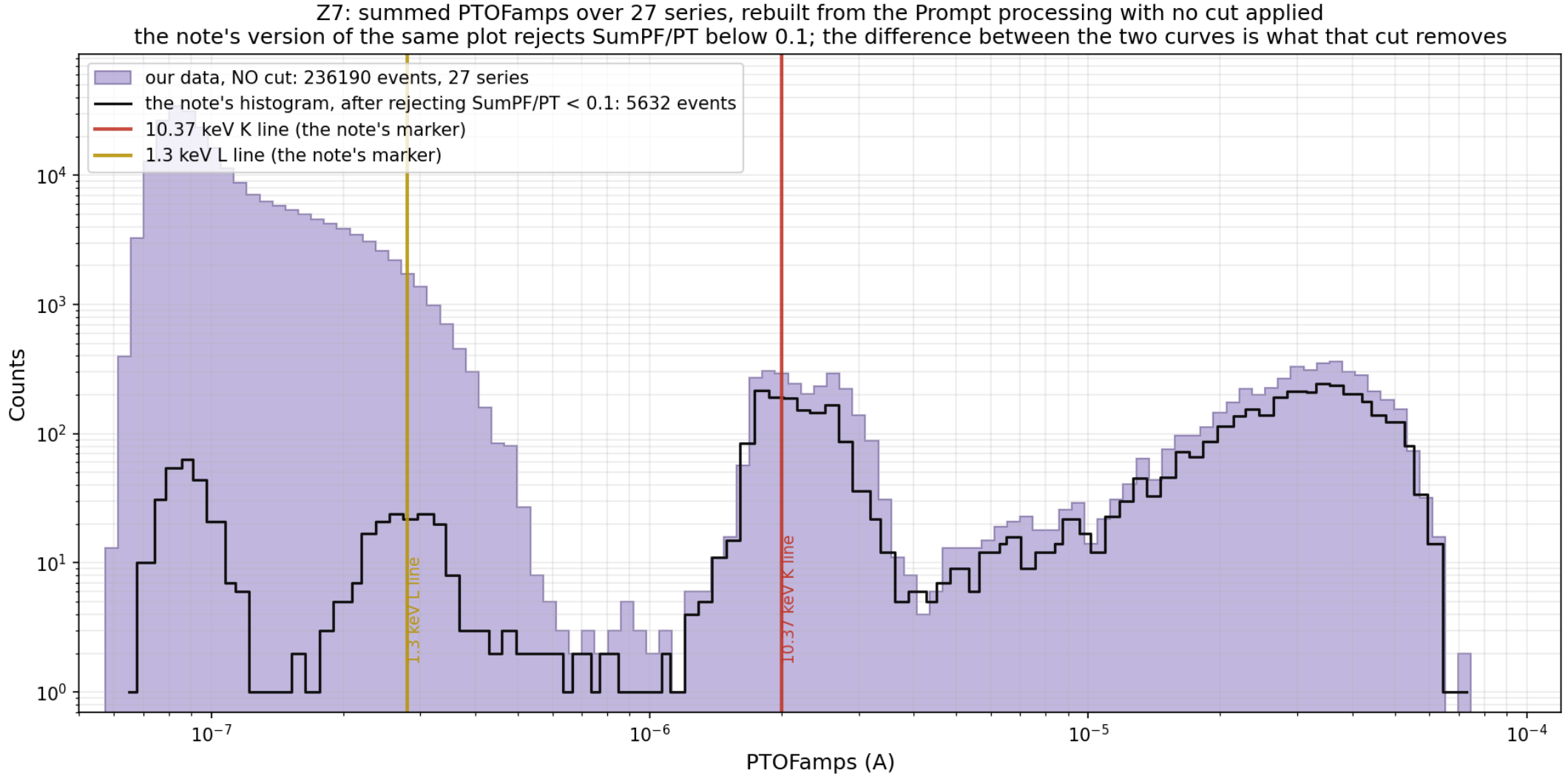
Z7 把一个 10.37 keV 事件的 $32 \pm 1\%$ 吸收进 TES。CDMS 文档在 R37 CUTE tower 上报的是 26–41%。

还没落实：只做了 Z7 · Z7 的 HV 偏压没有确认（按 0 V 算） · 没有 dI/dV，电感项被丢掉 · PDS2 就是那 $\pm 1\%$ · 12.5% 的事件在任何通道都拟合不上，没进直方图。

Backup — the events



Backup — 事件谱



Backup — the three versions of the power formula

$$\delta P = I_0(R_L - R_0) \cdot \delta I + c_2 \cdot R_L \cdot (\delta I)^2 \quad \text{with } c_2 = 2 \text{ (Method 1, used here), } 1 \text{ (exact small-signal), } -1 \text{ (Method 2)}$$

Relative to the exact form: Method 1 +0.58 %, Method 2 −1.15 % in energy — far below the 15 % single-channel width and the PDS2 bracket.

The inductor term of the full equations needs L and the loop gain from dI/dV, which do not exist for these series on MSI; it is neglected.

The official Eabs (trigger bin 16383, baseline bins 93..15758, 5-pole 20 kHz prefilter, window −0.5/+1 ms, same formula) is reproduced to 0.2 %; the grey outlines on the histogram slides are it.

Backup — 功率公式的三个版本

$$\delta P = I_0(R_L - R_0) \cdot \delta I + c_2 \cdot R_L \cdot (\delta I)^2 \quad c_2 = 2 \text{ (method 1 , 本报告用的)、 } 1 \text{ (精确的小信号解)、 } -1 \text{ (method 2)}$$

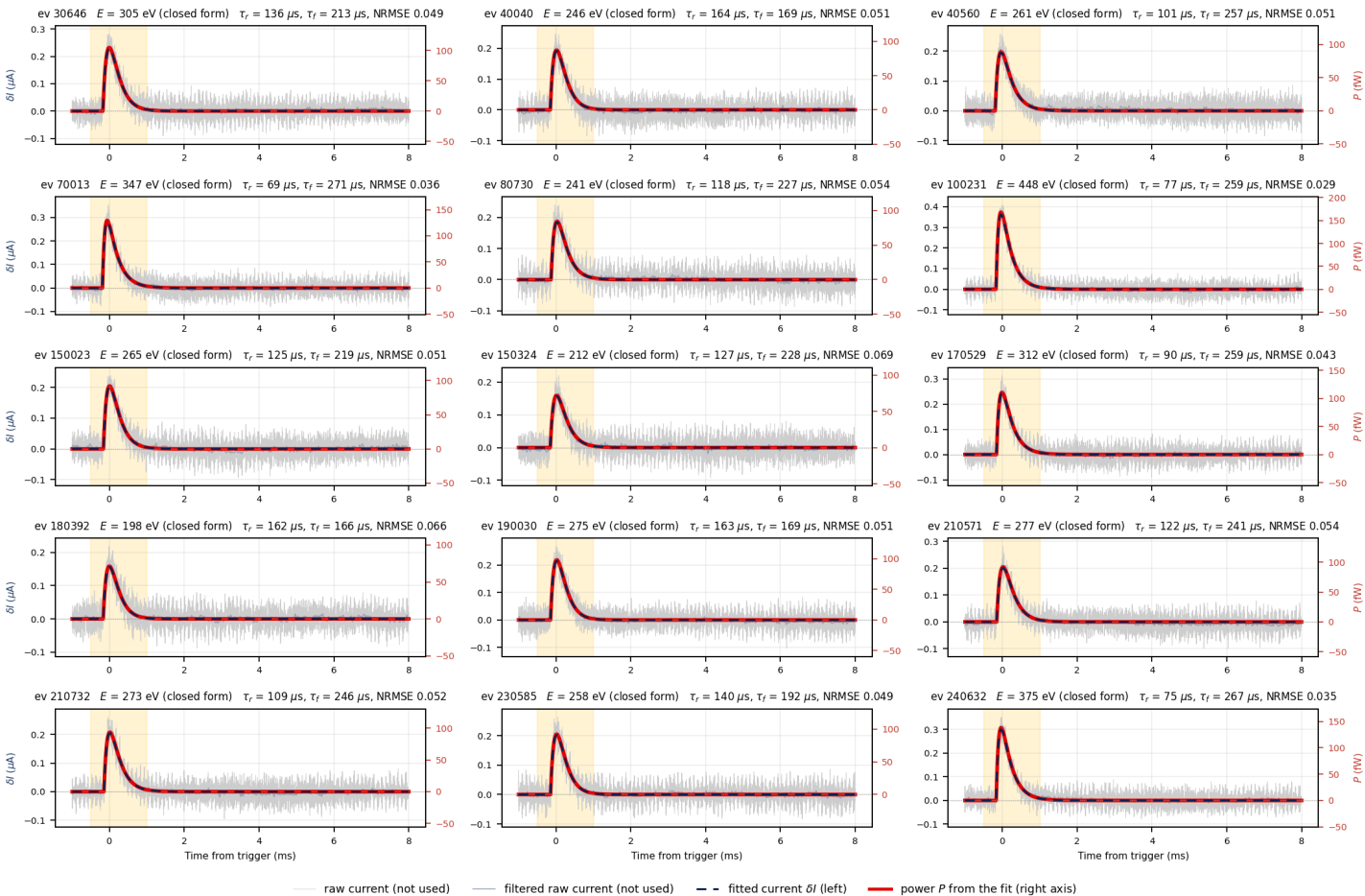
相对精确解：method 1 高 0.58% ，method 2 低 1.15% —— 远小于单通道 15% 的宽度和 PDS2 的范围。

完整方程里的电感项需要 L 和 loop gain ，要从 dI/dV 来，MSI 上这些 series 没有这个数据；这一项被忽略。

官方 Eabs （触发在第 16383 点，基线是 93..15758 点的平均，5 阶 20 kHz 预滤波，窗口 −0.5/+1 ms ，同一个公式）复现到 0.2% ；直方图页上的灰色轮廓就是它。

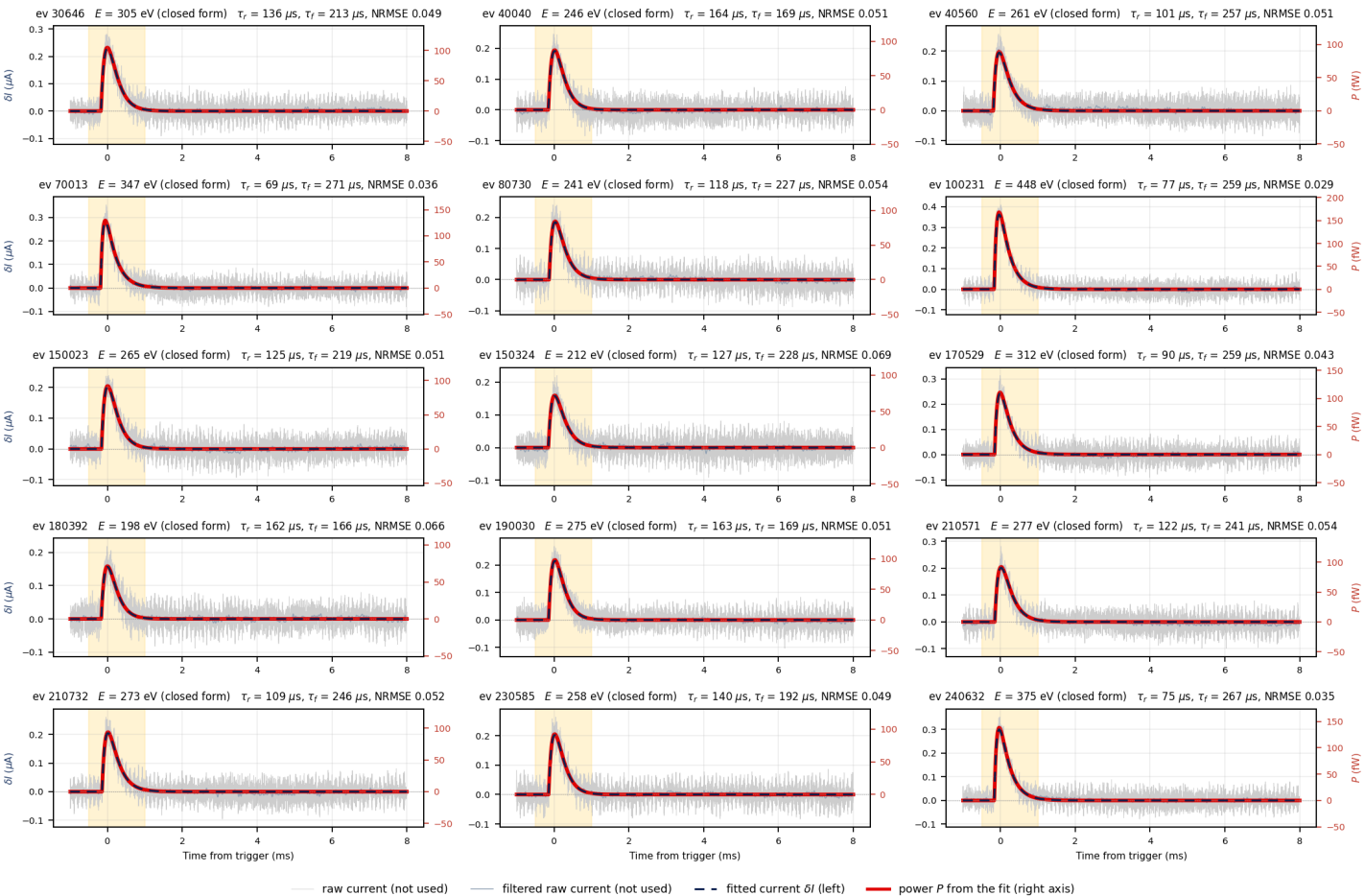
Backup — 15 events, fitted current and power (PBS1)

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)
Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * I^2 * d(t)^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_1) - \exp(-t/\tau_2)]$), 5 free params, pretrigger free in 16383+3000; the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



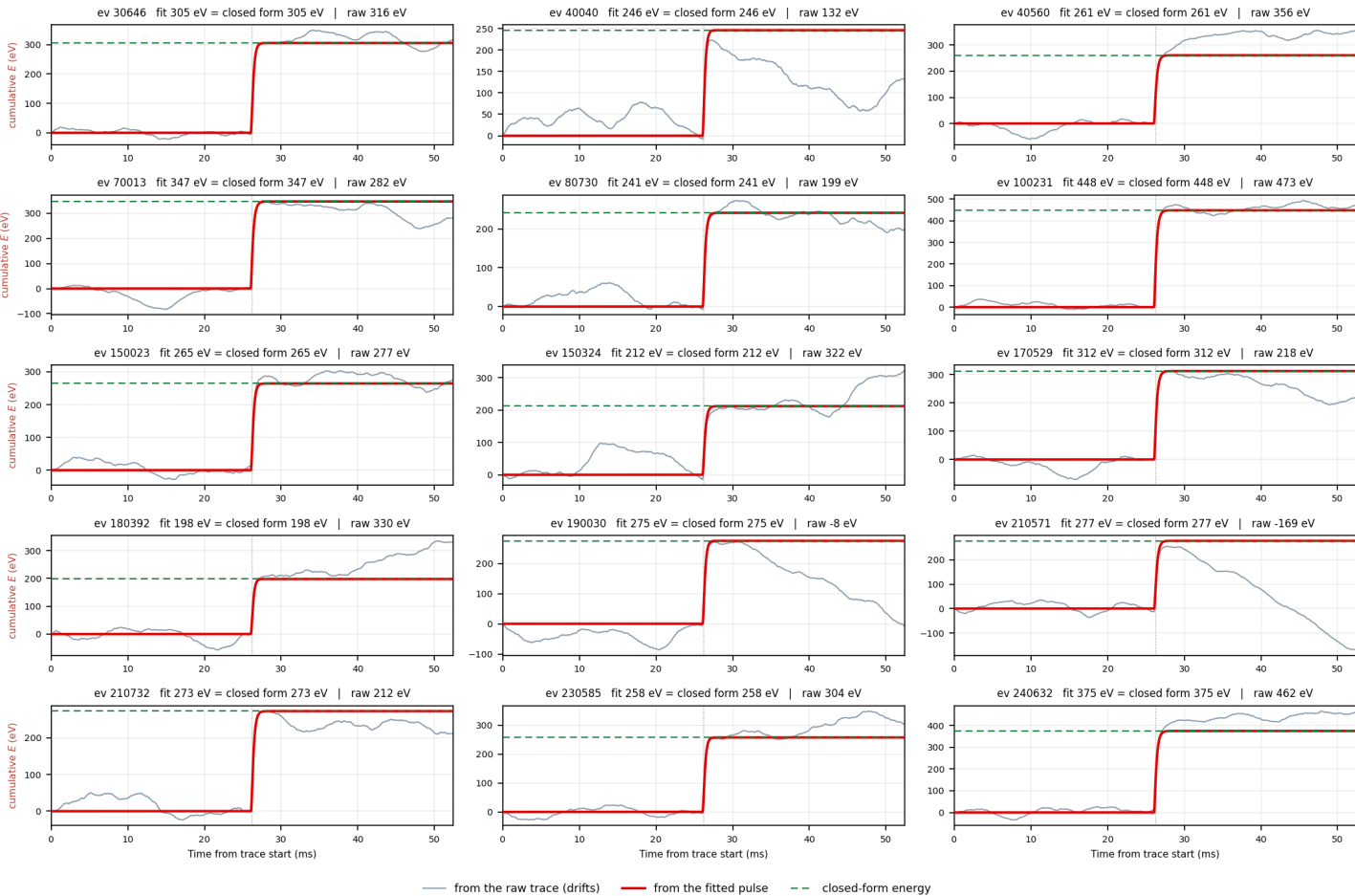
Backup — 15 个事件，拟合电流和功率（ PBS1 ）

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)
Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * I^2 * d(t)^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_1) - \exp(-t/\tau_2)]$), 5 free params, pretrigger free in 16383+3000; the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



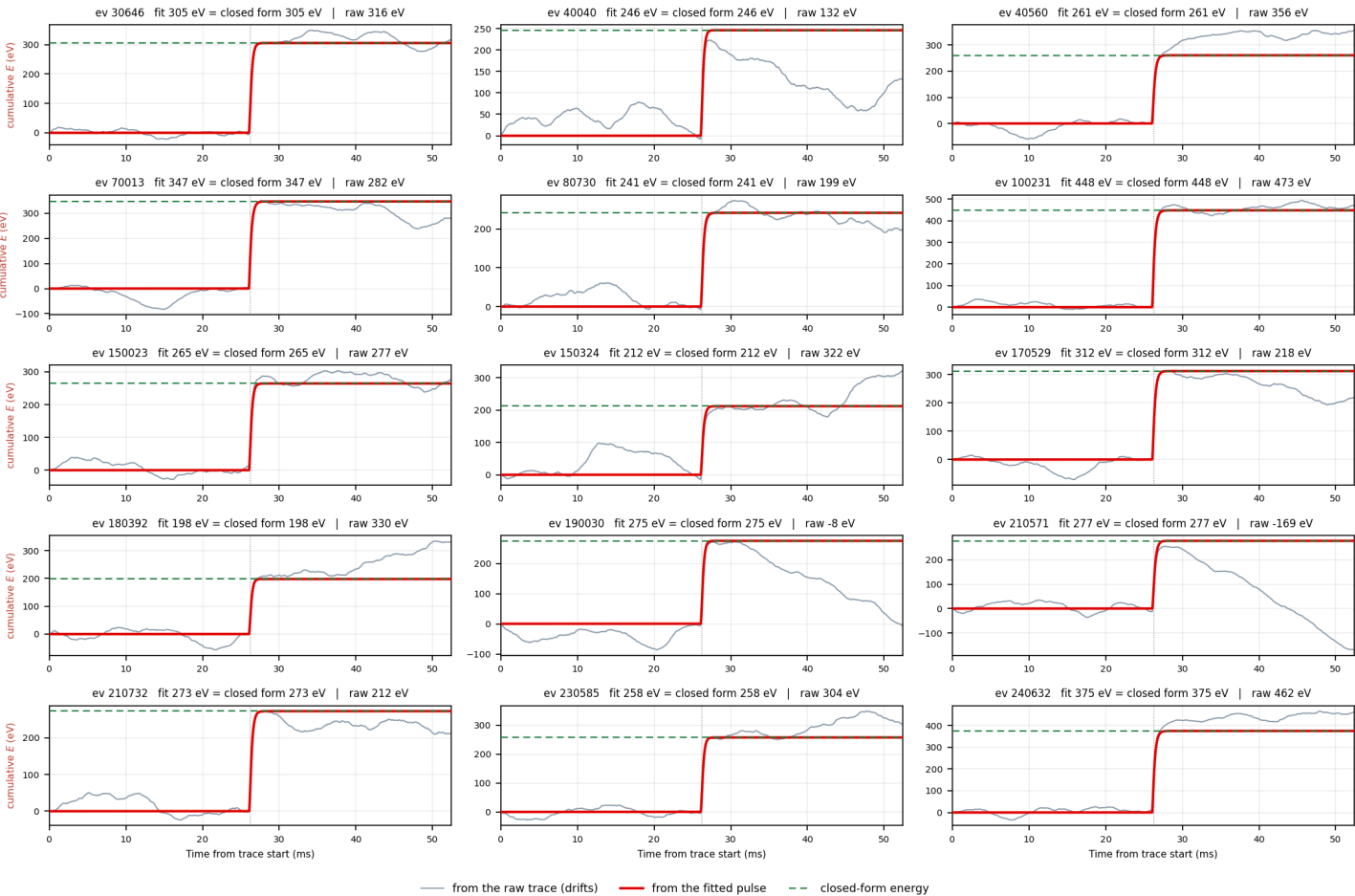
Backup — 15 events, cumulative energy (PBS1)

Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace
Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * (d(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_1) - \exp(-t/\tau_2)]$), 5 free params, pretrigger free in 16383+3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



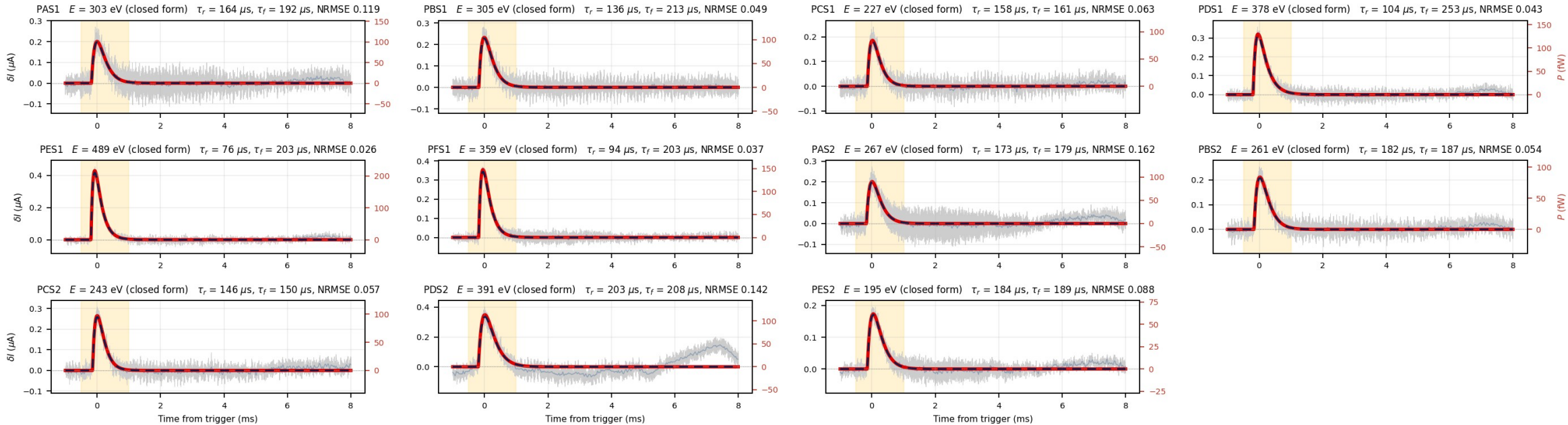
Backup — 15 个事件，累计能量（PBS1）

Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace
Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * (d(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
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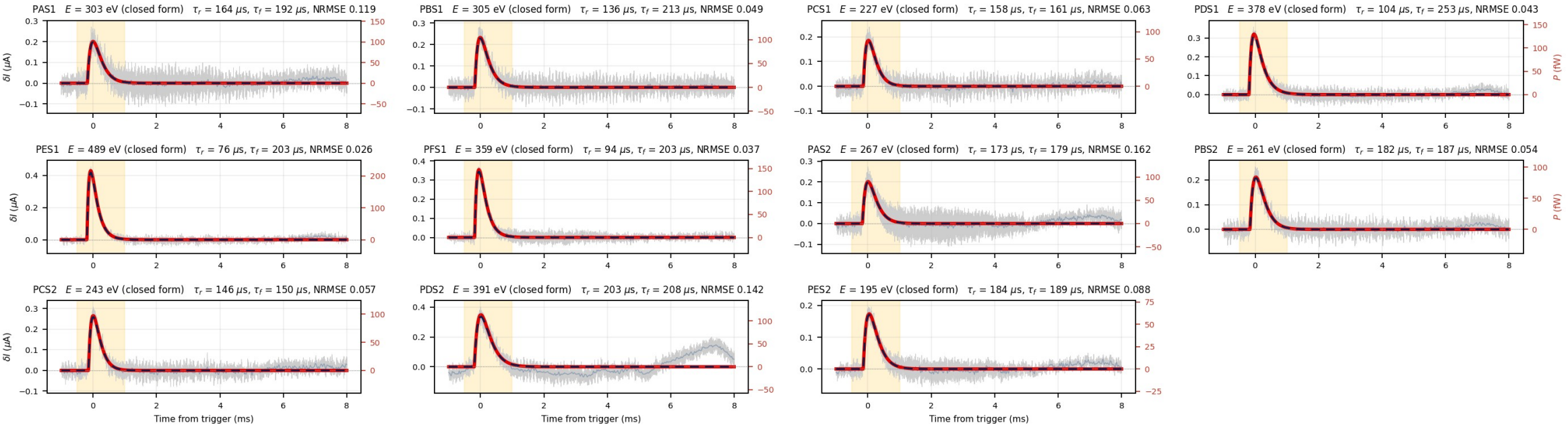
Backup — event 30646, all channels

Z7 event 30646, all channels: fitted current and power
sum over channels from the fits: 3419 eV = 33.0% of 10.37 keV (official-window raw sum 3397 eV = 32.8%)
Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * d(t)^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$, 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



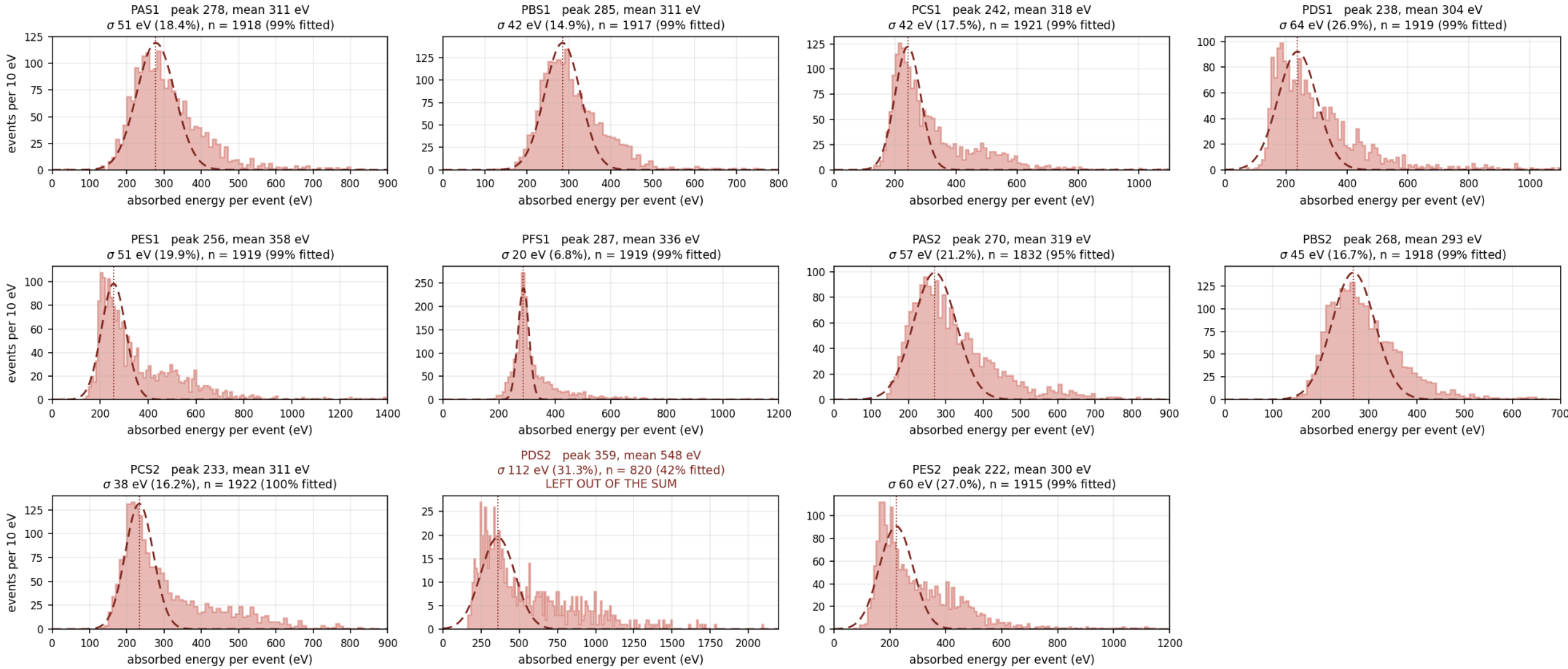
Backup — 事件 30646 , 全部通道

Z7 event 30646, all channels: fitted current and power
sum over channels from the fits: 3419 eV = 33.0% of 10.37 keV (official-window raw sum 3397 eV = 32.8%)
Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * d(t)^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$, 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — energy distribution of every channel

Z7: energy of every K-line event, one panel per channel (30 series)
each entry is one event: its trace fitted with the two-exponential and the power integrated in closed form, method1 quadratic
the dashed curve is a Gaussian fit to the core, so its centre is the PEAK; these distributions are right skewed, so peaks must not be added (the plain means do add, to 3710 eV = 35.8% of the 10.37 keV K line)



Backup — 每个通道的能量分布

Z7: energy of every K-line event, one panel per channel (30 series)
each entry is one event: its trace fitted with the two-exponential and the power integrated in closed form, method1 quadratic
the dashed curve is a Gaussian fit to the core, so its centre is the PEAK; these distributions are right skewed, so peaks must not be added (the plain means do add, to 3710 eV = 35.8% of the 10.37 keV K line)

